ACADEMIUM ENGINE PACKET

Section 3: Consequence Generation Rules

Version: 2.0 **Status:** Active Specification **Scope:** Applies to all consequence narratives generated after Turn 1 (MC), Turn 2 (FR), and Turn 3 (FR). Does not govern the Final Outcome narrative (Section 8 of case structure) or the optional Reflection prompt. **Inputs received from:** Section 2 (RDS band, extracted signals, competency evidence), Turn 1 tradeoff signature, prior decision history **Outputs sent to:** Section 4 (KPI Computation), Section 6 (Causal Explanation Generation), Section 7 (Response Assembly)

3.1 Purpose and Pedagogical Foundation

The consequence narrative is the engine's most critical output. It is the moment where the student's decision becomes real — where abstraction becomes experience. Everything else in the engine (signal extraction, KPI computation, causal explanations) exists to support this moment.

A consequence narrative has exactly one job: **to show the student what their decision produced in the world of the simulation, honestly and without judgment.**

It does not teach. It does not evaluate. It does not hint at what a better decision would have been. It reports — with the calm, factual precision of a senior colleague describing what happened in the organization after a strategic commitment was made.

Research Basis

Experiential Learning Theory (Kolb, 1984): The consequence narrative IS the Concrete Experience stage of Kolb's cycle. It must be vivid, specific, and realistic enough to trigger genuine reflective observation. A vague, sanitized, or generic consequence fails this stage entirely — the student cannot reflect on something that did not feel real.

Productive Failure (Kapur, 2008, 2016): Negative consequences are not failures of the simulation. They are the simulation working correctly. Research demonstrates that learners who encounter negative outcomes and reflect on them develop deeper understanding than learners who receive only positive reinforcement. The engine must generate honest consequences — including negative trade-offs — without framing them as punishments.

Psychological Safety (Edmondson, 1999): The narrative must never make the student feel judged, embarrassed, or penalized for their reasoning. The scene being described is something that happened to the organization — not a verdict on the student's intelligence or character. This

is the tension at the heart of consequence generation: consequences must be honest AND psychologically safe. Both simultaneously. Always.

Feedback Intervention Theory (Kluger & DeNisi, 1996): Feedback directed at the task and process improves learning. Feedback directed at the ego (praise or criticism of the person) impairs it. The consequence narrative must describe organizational events, not comment on the student's decision-making quality.

Self-Determination Theory (Deci & Ryan, 1985): Students must feel autonomous — that their choices had genuine, meaningful effects. A consequence narrative that is indistinguishable regardless of what the student chose destroys this sense of autonomy and breaks trust in the simulation. Every consequence must be traceable to the specific decision that produced it.

3.2 Input Requirements

The consequence generator must have access to all of the following before generating any narrative. If any input is missing, the pipeline halts and returns a graceful error — it does not generate a partial consequence.

Input	Source	Purpose
Student's extracted signals	Section 2	Determines what the student's reasoning emphasized; anchors narrative to student's specific intent
RDS band	Section 2	Governs narrative length and richness (see 3.4)
Turn number	System state	Determines whether compounding logic applies (Turn 2+) and whether synthesis framing applies (Turn 3)
Prior decision history	Session state	Enables stakeholder continuity and compounding effects
Turn 1 tradeoff signature	Case definition	Establishes the directional trajectory the simulation is on
Case consequence templates	Case definition	Pre-authored directional content; engine generates narrative around these, never contradicts them
Active KPIs for this case	Case configuration	Ensures narrative references only the metrics that are active in this simulation

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3.3 Narrative Structure: Four Required Elements

Every consequence narrative, regardless of RDS band, must contain these four elements. They may appear in any order, but all four must be present.

Element 1: Observable Outcome

What it is: A concrete description of something that changed in the organization as a result of the student's decision. Not an assessment of whether that change is good or bad — just a factual account of what happened.

Rules:

- Must be observable — something that a person inside the organization would see, hear, or measure
- Must be specific — names or implies a team, process, timeline, relationship, or metric
- Must be causal — the student's decision must be the clear antecedent
- Must NOT be evaluative — no language that implies whether this outcome is desirable or undesirable

Correct: *"Following the decision to renegotiate the delivery timeline, the operations team received a formal request to revise the Q3 project schedule."* **Incorrect:** *"The decision to renegotiate the timeline was the right call and led to a productive restructuring of the project schedule."* (Evaluative — "right call")

Element 2: Stakeholder Reaction

What it is: At least one named or clearly implied stakeholder responds to the decision in a way that reflects the organizational dynamics of the case.

Rules:

- Stakeholders must react consistently with who they were established to be in the case context
- Stakeholders have memory across turns — their reactions in Turn 2 must acknowledge what happened in Turn 1
- Reactions must be realistic — neither uniformly positive nor uniformly negative
- Reactions must not resolve the tension entirely — stakeholder dynamics should remain open, not closed

Correct: *"The client's project lead acknowledged the communication but indicated they would need to escalate the revised timeline to their own board before confirming the new arrangement."*

Incorrect: *"The client was very happy with the communication and immediately agreed to everything."* (Resolves tension; unrealistically clean)

Element 3: New Information

What it is: One piece of information the student did not have before this consequence — a development, a discovery, a secondary effect, or an emerging tension that the decision revealed.

Rules:

- Must be genuinely new — not a restatement of the case context
- Must be relevant to the ongoing challenge — not random noise
- Must not resolve the case — it should deepen complexity, not simplify it
- Follows the Half-Story Rule: consequences reveal part of the picture; the full picture is never provided

Correct: *"In a separate development, an internal audit flagged that the budget reallocation required to accommodate the new timeline would affect two other active projects, a consideration the operations team had not surfaced during initial planning."* **Incorrect:** *"Fortunately, it turned out there was extra budget available that made the solution easy to implement."* (Resolves tension artificially; uses prohibited word "fortunately")

Element 4: Forward Implication

What it is: A thread that connects the current consequence to the next decision point. Not a clue about what the right answer is — a realistic signal that the situation is ongoing and the next decision matters.

Rules:

- Applies to Turn 1 and Turn 2 consequences only. Turn 3 consequences do not include a forward implication (they feed into the Final Outcome narrative)
- Must not direct the student toward a specific choice
- Must create genuine tension — a situation where the student has a real decision to make, not a situation that has been conveniently set up for a single correct response

Correct: *"The team is now waiting for direction on how to prioritize competing commitments before the end of the week."* **Incorrect:** *"The team is waiting, and the only way to resolve this is to immediately communicate a clear priority to all stakeholders."* (Directive — implies a specific action)

3.4 RDS-Governed Narrative Richness

The RDS band received from Section 2 directly governs the depth and length of the consequence narrative. This is the mechanism by which the engine proportionally rewards more sophisticated reasoning with more nuanced outcomes — without ever grading or evaluating the student.

Surface Band (RDS 1–4)

Parameter	Specification
Target length	80–100 words
Observable outcomes	1–2
Stakeholder reactions	1 (primary stakeholder only)
New information pieces	1
Compounding visibility	Minimal — brief acknowledgment of prior turn if Turn 2+
Forward implication	Simple and direct
Tone	Factual, brief, direct

Pedagogical rationale: A student in the Surface band has given minimal reasoning. Their consequence mirrors that: the world responded, but the response is contained and direct. This is not a punishment — it is a proportional reflection of the decision they engaged with. Simpler reasoning produces a simpler consequence landscape.

Engaged Band (RDS 5–9)

Parameter	Specification
Target length	100–130 words
Observable outcomes	2–3
Stakeholder reactions	1–2 (primary + at least one secondary stakeholder)
New information pieces	1–2
Compounding visibility	Moderate — clearly references the trajectory from prior turns
Forward implication	Introduces a complication or tension

Parameter	Specification
Tone	Factual, measured, acknowledges organizational complexity

Integrated Band (RDS 10–15)

Parameter	Specification
Target length	130–160 words
Observable outcomes	3+ with visible compounding between them
Stakeholder reactions	2+ including at least one secondary or unexpected reaction
New information pieces	2+
Compounding visibility	Full — narrative explicitly builds on the cumulative decision trajectory
Forward implication	Surfaces a genuine dilemma with multiple defensible responses
Tone	Factual, nuanced, reflects organizational systems and interconnections

Pedagogical rationale: A student in the Integrated band demonstrated multi-signal reasoning. Their consequence reflects that: the world responded in kind — with complexity, second-order effects, and stakeholder dynamics that require continued engagement. This mirrors the real experience of sophisticated decision-making: the more thoughtfully you engage, the more you see how interconnected everything is.

3.5 Compounding Logic

Consequences do not exist in isolation. From the second turn onward, the consequence narrative must reflect the cumulative trajectory the student has created across all prior decisions. This is what makes the simulation feel like a coherent organizational story rather than a series of disconnected episodes.

This logic applies regardless of whether the professor configured a 4-step or 10-step case. The engine always operates in positional terms — **First**, **Intermediate**, or **Final** — not by absolute turn number.

Rules for Compounding

Rule 1: Stakeholders have memory. If a stakeholder expressed frustration, concern, or a specific

position in a prior consequence, that position must still be present in the current consequence unless a specific, narratively supported event changed it. Stakeholders do not reset between turns. This applies across all turns in a case, whether there are 4 or 10.

Rule 2: Metrics accumulate. If a KPI has already been stressed in a prior turn, the current consequence must acknowledge the accumulated pressure — not treat the situation as fresh. A team under morale pressure since Turn 1 is more fragile in Turn 6 than a team encountering stress for the first time. The longer the case, the more accumulated pressure matters.

Rule 3: Strategic posture compounds. The student's first MCQ selection established a directional trajectory (conservative, aggressive, stakeholder-first, efficiency-focused). All subsequent FR consequences must reflect whether the student's responses are consistent with or in tension with that initial posture. Tension between postures is a valid and realistic narrative element — it should be surfaced, not smoothed over. In longer cases (7–10 turns), posture drift across multiple turns becomes a meaningful narrative thread.

Rule 4: The full case history is always available. The consequence generator must receive the full session state at every turn. It cannot generate contextually accurate consequences from the current turn's input alone. In a 10-turn case, that means the engine has access to 9 prior decisions and their consequences before generating the 10th.

Compounding Decision Tree

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IF position = FIRST (always MCQ, always Turn 1 regardless of total step count)
  → Generate consequence from tradeoff signature only
  → No compounding required
  → Include forward implication

IF position = INTERMEDIATE (any turn between First and Final)
  → Generate consequence from current signals + full prior history
  → Reference at least one element from the immediately prior consequence
  → In cases of 7+ turns, also reference a pattern across earlier turns
    if one has emerged (e.g., consistent stakeholder tension, recurring metric
pressure)
  → Check stakeholder continuity against all prior turns
  → Compound metrics if prior turns created KPI stress
  → Include forward implication

IF position = FINAL (last turn regardless of total step count)
  → Generate consequence from final signals + full session history
  → Reference the cumulative trajectory across all turns, not just the most recent
  → Surface the tension or coherence between the student's choices across the full
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case
  → DO NOT include forward implication (feeds into Final Outcome narrative)
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Scaffolding Progression Across Step Counts

The professor's step count configuration also governs the scaffolding progression. The engine must adjust its narrative register accordingly:

Position in Case	Scaffolding Level	Narrative Implication
First 25% of turns	High scaffolding	Consequences are clear and direct; second-order effects are limited; stakeholder reactions are straightforward
Middle 50% of turns	Medium scaffolding	Consequences begin to compound; secondary stakeholder reactions emerge; tensions become less resolved
Final 25% of turns	Low scaffolding	Consequences are complex and multi-layered; compounding effects are fully visible; tensions are unresolved and require synthesis

This means a 4-turn case reaches full complexity at Turn 3, while a 10-turn case reaches it at Turn 8. The engine calibrates richness to position, not to absolute turn number.

3.6 The Tradeoff Configuration System

Tradeoffs in Academium are professor-configurable. The engine must check the professor's tradeoff configuration before generating any consequence narrative and apply the appropriate behavior for each of the three possible states.

The Three Tradeoff Configuration States

State A: Professor provided specific tradeoff text The professor wrote a custom tradeoff in a text box during case setup (e.g., "Prioritizing client satisfaction will always strain internal team capacity in this scenario"). The engine must anchor the consequence narrative to this specific tradeoff. It must surface, in some form, the tension the professor defined. The exact wording does not need to be reproduced, but the directional content must be present.

State B: Tradeoffs enabled, no specific text provided The professor enabled tradeoffs but did not define custom content. The engine generates realistic tradeoffs naturally, derived from the student's decision signals and the case's KPI dynamics. The engine uses the criteria below to ensure the tradeoff is substantive, not formulaic.

State C: Tradeoffs not configured (disabled or left blank) The professor did not configure tradeoffs for this case. The engine does not force a tradeoff element. However, the engine must still generate realistic consequences — it must not produce artificially rosy outcomes that imply a decision had no costs. Natural organizational tensions may still emerge from the simulation dynamics; they are simply not required.

What Counts as a Substantive Tradeoff (States A and B)

Qualifies:

- A stakeholder who benefits less than another (*"While the client relationship stabilized, the operations team faced renewed pressure on their delivery commitments"*)
- A metric that moved in the opposite direction (*"The decision maintained stakeholder trust but created short-term budget strain"*)
- An unintended secondary effect (*"The communication strategy resolved the immediate tension but surfaced a longer-term misalignment between client expectations and internal capacity"*)
- A cost that is delayed, not eliminated (*"The renegotiated timeline bought time, but the underlying resource constraint remains unresolved"*)

Does not qualify:

- A generic disclaimer (*"As with all decisions, there are trade-offs"*) — formulaic, not substantive
- A trivial inconvenience that doesn't connect to any active KPI or stakeholder
- A future risk so vague it creates no tension (*"Time will tell how this plays out"*)

Critical Rule Across All States

Even when tradeoffs are not configured (State C), the engine must never generate a consequence that implies a decision was cost-free or universally positive. The difference between State C and States A/B is not "tradeoffs disappear" — it is "the engine does not enforce them as a structural requirement." Realistic organizational dynamics will naturally produce tensions. The engine simply does not add them artificially when the professor has not configured them.

3.7 Prohibited Language: Complete List

The following words, phrases, and constructions are absolutely prohibited in consequence narratives. There are no exceptions.

Evaluative Language (Implies a right/wrong answer)

Prohibited	Why
"Correct," "incorrect"	Implies a right answer exists
"Right," "wrong" (as judgment)	Same
"Best," "optimal," "ideal"	Implies a superior path
"Good decision," "poor decision"	Direct evaluation
"Well done," "good job"	Praise violates non-graded principle
"You should have," "you could have"	Retroactive judgment
"The better choice would have been"	Answer-giving
"As expected"	Implies predictable correct answer
"Predictably" (when used judgmentally)	Same
"Perfect"	Implies a ceiling was reached
"Mistake," "error" (applied to decision)	Moral judgment on a trade-off

Emotionally Loaded Language (Violates neutral tone)

Prohibited	Why
"Unfortunately"	Implies regret about the outcome
"Fortunately," "luckily"	Implies the outcome was an unearned positive
"Sadly"	Loaded emotional framing
"Worryingly"	Loaded; also evaluative
"Surprisingly," "unexpectedly"	Implies a predictable path existed
"Disappointingly"	Evaluative and emotionally loaded
Exclamation points (!)	Performative; violates mentorship tone

Answer-Giving Patterns (Prohibited regardless of phrasing)

The engine must never — directly, indirectly, or by strong implication — communicate that a different decision would have produced a better overall outcome. This includes:

- Phrasing that makes one path's consequences sound clearly preferable to another
- Narrative framing that sets up the next decision with an implied "correct" response
- Language that describes current consequences as "avoidable" without also acknowledging what was gained by the choice made
- Framing that presents the student's decision as surprising or unusual in a way that signals it was suboptimal

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3.8 Required Language Patterns

The following constructions are explicitly approved for consequence narrative use. They are not mandatory templates — they are safe anchors the engine can use when generating organizational language.

Cause-effect connectors:

- "This led to..."
- "As a result..."
- "In response to this decision..."
- "Following the commitment to [action]..."
- "This created..."
- "The decision set in motion..."

Stakeholder reaction language:

- "The [stakeholder] responded by..."
- "Within the [team/organization], the reaction was..."
- "The [stakeholder] indicated that..."
- "From the client's perspective..."
- "The [team/group] noted that..."
- "This prompted [stakeholder] to..."

Neutral consequence language:

- "The situation shifted..."
- "A new dynamic emerged..."
- "An unresolved tension became more visible..."
- "The organization found itself navigating..."
- "An opportunity emerged alongside a new constraint..."
- "This created pressure on..."

New information framing:

- "A separate development revealed..."
- "In parallel, the [team/stakeholder] surfaced..."
- "What had not been fully visible before became clear..."
- "The decision exposed..."

3.9 Tone Specification: The Senior Colleague Standard

The consequence narrative is written from the perspective of a senior colleague who has spent 20 years in the industry observing organizational dynamics. They are not lecturing. They are not cheerleading. They are not warning. They are describing — with the calm authority of someone who has seen how these situations unfold and trusts the listener to draw their own conclusions.

The Four Tone Tests

Before any consequence narrative is considered complete, it must pass all four tests:

Test 1 — The Lecture Test: Read the narrative and ask: "Does this sound like a professor explaining a lesson, or like a scene in an organizational story?" If it reads like a lesson, it has failed. Rewrite as a scene.

Test 2 — The Praise Test: Read the narrative and ask: "Would a student reading this feel evaluated — either positively or negatively?" If yes, remove or rephrase the evaluative elements.

Test 3 — The Hint Test: Read the narrative and ask: "Does anything in this narrative make one particular response to the next decision prompt seem more clearly correct than others?" If yes, the narrative has given an answer. Remove or rephrase.

Test 4 — The Reality Test: Read the narrative and ask: "Could this realistically have happened in an actual organization, with actual people, in response to this actual decision?" If the outcome seems contrived, implausible, or engineered to create a specific lesson, revise for realism.

3.10 Regeneration Rules

Students may regenerate a consequence narrative once per turn. Regeneration does not change what happened — it changes how it is described.

What Regeneration Changes

- The specific scene used to describe the outcome (a different meeting, a different stakeholder interaction, a different day in the organization's timeline)
- The secondary details and examples
- The specific stakeholder quoted or described

What Regeneration Does NOT Change

- The directional outcome (if the consequence involved a KPI decrease, that decrease still occurred)
- The metric computations (already locked when the original consequence was delivered)
- The fundamental trade-off (the negative element must still appear in the regenerated version)
- The causal connection to the student's decision

Regeneration Failure Mode to Prevent

A regenerated consequence that is noticeably "better" — more positive, more flattering, more validating — than the original implies that the first consequence was somehow a penalty. Both versions must feel like neutral organizational descriptions of the same reality, told from a different vantage point.

3.11 Quality Gates

Before any consequence narrative is delivered to the student, it must pass all seven quality gates. If any gate fails, the narrative is regenerated — not delivered in partial or flagged form.

Gate	Check	Pass Condition
G1: Prohibited Language	Scan for all terms in Section 3.7	Zero prohibited terms present
G2: Answer-Giving	Does the narrative imply a different decision would have been clearly better?	No directional hint toward a specific next response
G3: Causal Traceability	Can every narrative element be traced to the student's decision or prior turn history?	All elements causally grounded
G4: Tradeoff Configuration	What is the professor's tradeoff configuration state?	State A: Specific professor text is anchored in narrative. State B: Engine-generated substantive tradeoff is present. State C: No artificial tradeoff forced; narrative remains realistic.
G5: Stakeholder Continuity	Are stakeholders behaving consistently with their established positions from prior turns?	No unexplained stakeholder reversals
G6: Four Elements Present	Does the narrative contain all four required elements (Observable Outcome, Stakeholder Reaction, New Information, Forward Implication)?	All four present (Forward Implication required only for Turns 1–2)
G7: Tone	Does the narrative pass all four Tone Tests?	Passes Lecture Test, Praise Test, Hint Test, Reality Test

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3.12 Worked Examples

All examples in Spanish (Case 001). English translations in brackets.

EXAMPLE A — Turn 2 Consequence, Surface Band (RDS 1–4)

Context: Student chose Option B in Turn 1 (stakeholder-first posture). Turn 2 input was *"Hablaría con el cliente."* (RDS = 3, Surface Band)

Generated consequence:

"El equipo de gestión de cuentas programó una llamada con el cliente para esa semana. En la conversación, el cliente expresó preocupación por los cambios en el cronograma y solicitó una actualización formal por escrito antes del viernes. El equipo interno, por su parte, aún no había recibido instrucciones claras sobre cómo priorizar los proyectos activos."

["The account management team scheduled a call with the client for that week. In the conversation, the client expressed concern about the timeline changes and requested a formal written update before Friday. The internal team, for their part, had not yet received clear instructions on how to prioritize active projects."]

Quality gate check:

- G1 ✓ No prohibited language
- G2 ✓ No hint at what to do next
- G3 ✓ Client communication traces to student's decision
- G4 ✓ Tradeoff present (internal team without direction)
- G5 ✓ Client concern consistent with Turn 1 stakeholder dynamics
- G6 ✓ All four elements present
- G7 ✓ Passes all four tone tests

EXAMPLE B — Turn 2 Consequence, Integrated Band (RDS 10–15)

Context: Same Turn 1 selection. Turn 2 input demonstrated all five signals at STRONG/PRESENT. (RDS = 14, Integrated Band)

Generated consequence:

"La decisión de establecer comunicación directa y transparente con el cliente tuvo un efecto inmediato en la dinámica del proyecto. El cliente reconoció la actualización y solicitó que se formalizara en un addendum al contrato, lo que implicó involucrar al equipo legal más pronto de lo previsto. Al mismo tiempo, al revisar el cronograma revisado, el equipo de operaciones identificó un conflicto con otro proyecto prioritario que no había sido visible hasta entonces. La directora de operaciones solicitó una reunión urgente con el equipo antes de que cualquier compromiso fuera confirmado al cliente. Esta dinámica dejó al equipo en una posición en la que dos compromisos externos están en tensión directa, y la resolución de uno podría complicar el otro."

["The decision to establish direct and transparent communication with the client had an immediate effect on the project dynamics. The client acknowledged the update and requested it

be formalized in a contract addendum, which meant involving the legal team sooner than anticipated. At the same time, when reviewing the revised timeline, the operations team identified a conflict with another priority project that had not been visible until that point. The operations director requested an urgent meeting with the team before any commitment was confirmed to the client. This dynamic left the team in a position where two external commitments are in direct tension, and resolving one could complicate the other."]

Quality gate check:

- G1 ✓ No prohibited language
- G2 ✓ No answer-giving — two competing priorities, neither clearly superior
- G3 ✓ Transparency decision directly produced contract addendum request; operations conflict emerged from timeline revision
- G4 ✓ Multiple genuine tradeoffs: legal involvement, operations conflict, competing commitments
- G5 ✓ Client concern consistent with prior turns; operations team engaged as established
- G6 ✓ All four elements present
- G7 ✓ Passes all tone tests

EXAMPLE C — PROHIBITED Consequence (What NOT to Generate)

The following consequence is shown as a negative example. It must never be generated.

"¡Excelente decisión al comunicarte con el cliente! Esta fue claramente la mejor opción en esta situación. Gracias a tu elección, el cliente quedó satisfecho y la situación se resolvió de manera óptima. Si hubieras esperado más tiempo, los resultados habrían sido mucho peores. El equipo ahora puede avanzar sin problemas."

["Excellent decision to communicate with the client! This was clearly the best option in this situation. Thanks to your choice, the client was satisfied and the situation was resolved optimally. If you had waited longer, the results would have been much worse. The team can now move forward smoothly."]

Violations:

- "¡Excelente decisión!" — evaluative praise
- "claramente la mejor opción" — implies a correct answer
- "óptima" — prohibited word
- "Si hubieras esperado más tiempo, los resultados habrían sido mucho peores" — retroactive judgment; answer-giving

- No tradeoff present (G4 fails)
- Exclamation point in consequence narrative
- Resolves tension entirely — no forward implication of meaning

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3.13 Section Summary: What the Engine Must Do

At every decision turn, the consequence generator must:

1. **Receive all required inputs** before generating. Never generate from partial context.
2. **Apply the four required elements** (Observable Outcome, Stakeholder Reaction, New Information, Forward Implication for Turns 1–2).
3. **Scale narrative richness to the RDS band** received from Section 2.
4. **Apply compounding logic** for Turn 2 and Turn 3 (stakeholder memory, metric accumulation, posture consistency).
5. **Include at least one genuine tradeoff** in every narrative. No exceptions.
6. **Scan and pass all seven quality gates** before delivery. Regenerate if any gate fails.
7. **Pass all four tone tests** (Lecture, Praise, Hint, Reality).
8. **Apply regeneration rules** when the student requests an alternative framing — different scene, same trajectory, same tradeoff.

Section 3 connects to: Section 2 (receives RDS band and signal data), Section 4 (KPI Computation runs in parallel — consequence narrative and metrics are generated from the same inputs but independently), Section 5 (Metric Computation), Section 6 (Causal Explanation Generation — uses consequence narrative content to anchor "Why did this change?" explanations), Section 7 (Response Assembly — combines narrative, metrics, and causal explanations into final student-facing output).